DEEPSPACE GMSK DEMODULATION THROUGH SOLAR-WIND PLASMA SCINTILLATION

Replicating and Improving on Zhu et al. - Radio Science 2023

Part of ECE205 – Electromagnetic Engineering Project

SOLAR PLASMA CORRUPTS DEEP-SPACE RADIO LINKS

- **The Superior Conjunction Problem:** When the Sun-Earth-Probe angle drops below 5°, telemetry signals are forced through the solar corona.

- **The Electromagnetic Reality:** The corona is not empty space; it is a turbulent, high-energy plasma acting as a random, time-varying dielectric medium.

- **The Signal Degradation:** This non-uniform medium shreds the passing radio waves, causing severe amplitude scintillation and phase fluctuation.

- **The Consequence:** Massive spikes in Bit Error Rate (BER) and potential loss of critical spacecraft telemetry.

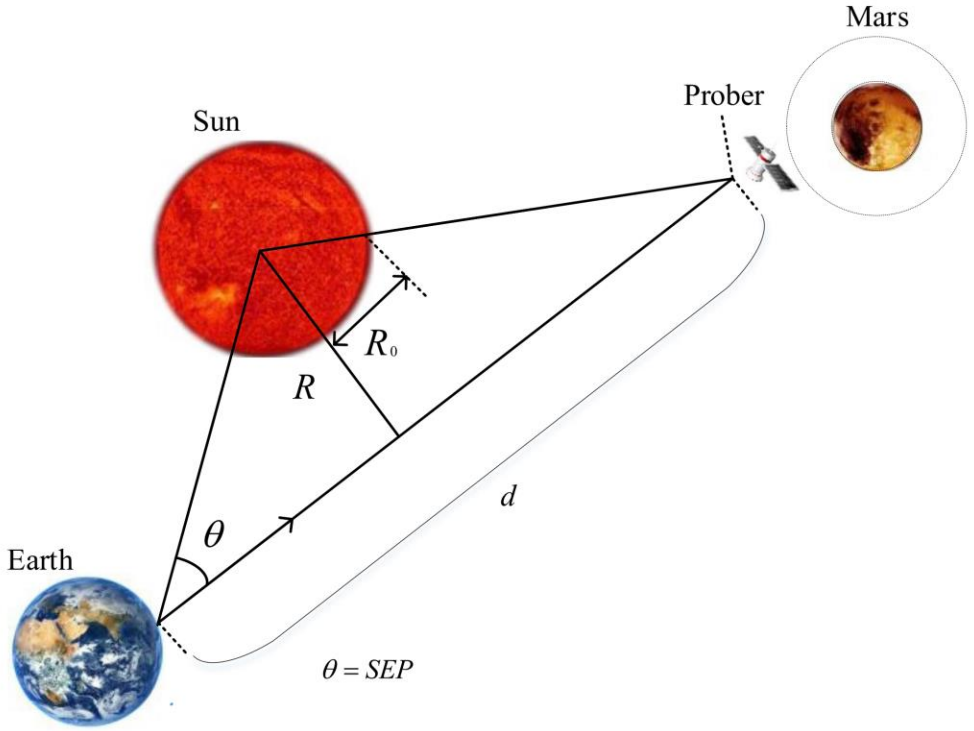


Figure 1 : Superior Solar Conjunction

*Geometry as per Zhu et al. 2023, Radio Science - https://doi.org/10.1029/2022RS007438Digital Object Identifier (DOI)

K-DISTRIBUTION CAPTURES PLASMA FADING STATISTICS

Received Signal Model (***r(t)***)

$$r(t) = \sqrt{p}h(t)x(t) + n(t)$$

- ***p*** : Power of the deep space prober transmitter
- ***x(t)*** : Transmitted signal, where x(t) ∈ {0, 1}
- ***h(t)*** : Turbulence channel gain coefficient (K-distributed for strong fading)
- ***n(t)*** : Additive White Gaussian Noise (AWGN), n(t) ~ N(0, σ^2)

Channel Parameter Link (***α***)

$$\alpha = \frac{2}{m-1}$$

❖ The K-distribution parameter ***α*** is determined by the scintillation index ***m***

Scintillation Index (m)

$$m = \frac{\langle I^2 - \langle I \rangle^2 \rangle}{\langle I \rangle^2}$$

- ***I*** : Received signal strength (the square of the received signal amplitude)
- (.): The ensemble average

❖ **Regimes** :
M < 1 (Weak Scintillation), Rician Fits
M ≥ 1 (Strong Scintillation) , h(t) ~ K-Distribution

❖ **Test conditions in this work** :
M = 1.2 (Mild Scintillation) / M = 1.4 (Heavy Scintillation)

*Channel Modelling is not our contribution. We use Zhu's K-distribution model as published, with their exact M value. Receiver design is what we IMPROVED!

OUR BASELINE: ZHU ET AL'S 1D-CNN + BI-LSTM DEMODULATOR

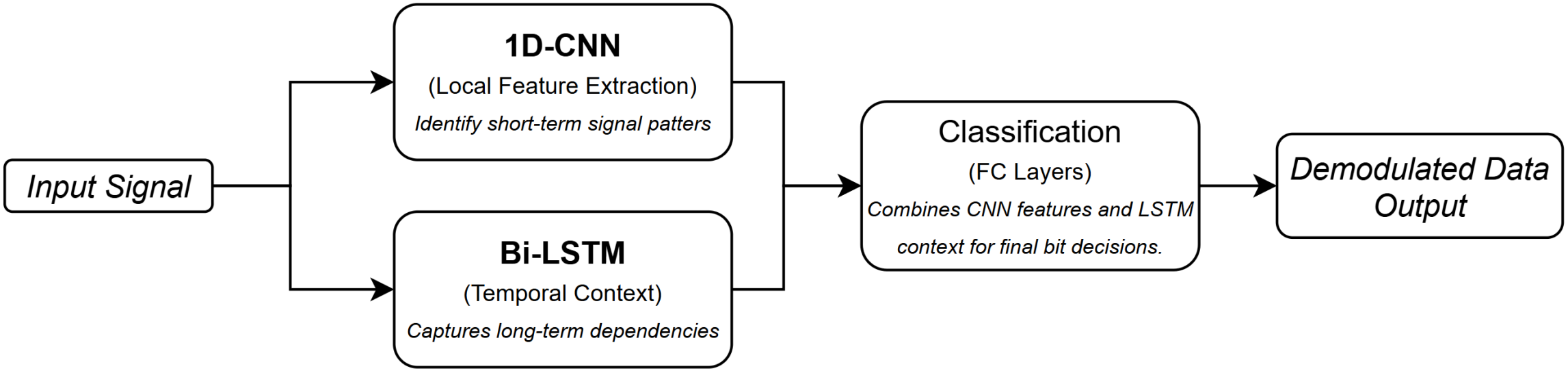


Figure 2 : Zhu's Model Architecture Diagram

Two parallel branches extract complementary features, then a classifier head decides each bit.

1D- CNN Branch

- Slides an 11-sample window across the I/Q waveform
- Catches local patterns - pulse shape, edges
- "What does this short snippet look like?"

Bi-LSTM Branch

- Reads the symbol stream forward AND backward
- Uses sequence context to disambiguate noisy bits
- "What does this bit's neighborhood tell me?"

*Architecture figure as per Zhu et al. 2023, Radio Science - doi:10.1029/2022RS007438

WE REPRODUCED IT: 3.122% BER ON THE PUBLISHED TEST SET

- Re-implemented Zhu's architecture in PyTorch from the paper
- Trained on Zhu's published Zenodo dataset (42K samples)
- Tested on Zhu's held-out test set (4,200 frames, 6 conditions)

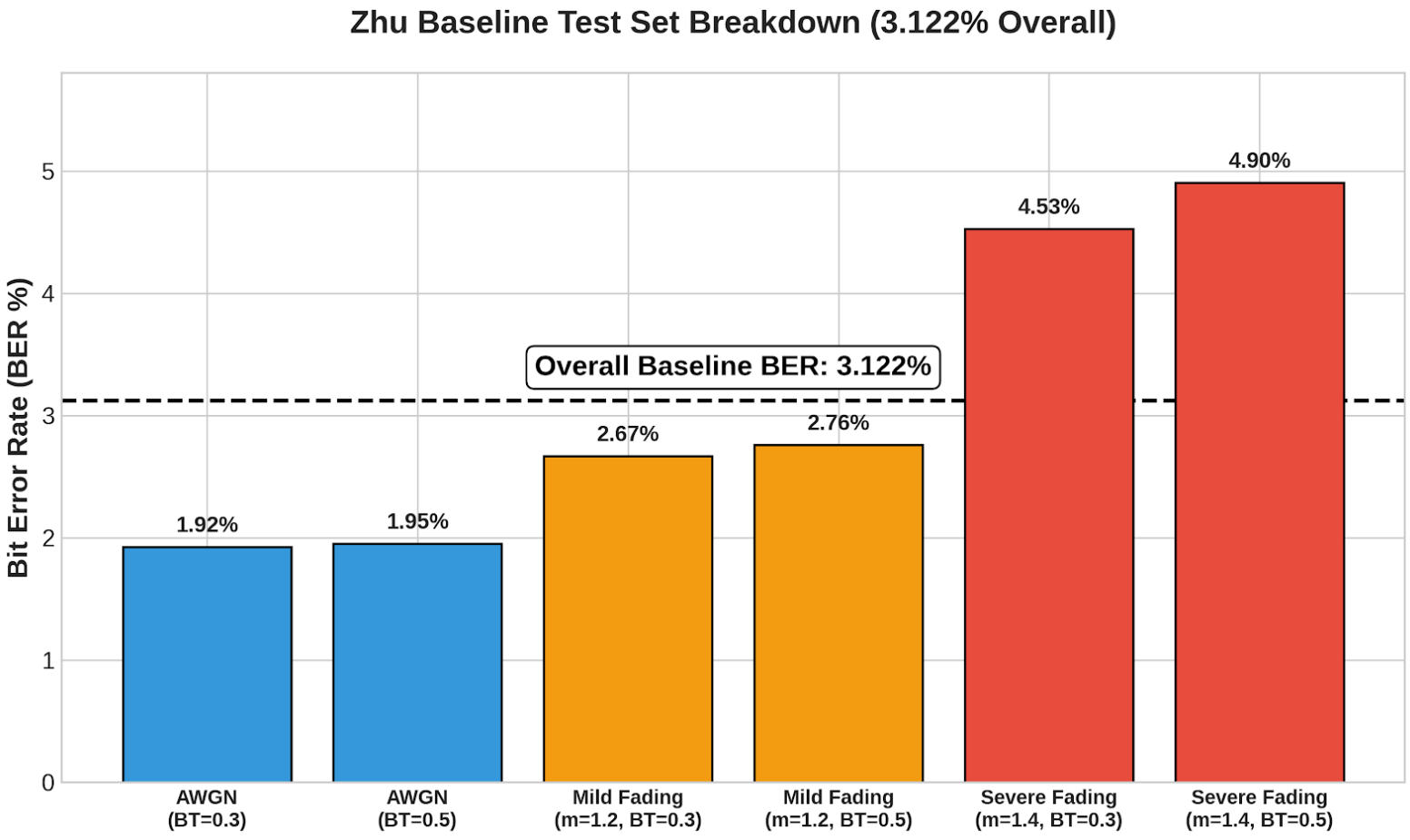


Figure 3 : Zhu Baseline Test-set Breakdown

Note: Zhu paper cites 63K training samples; Zenodo only has 42K*,
Zhu's paper itself reports BER-versus-SNR curves only, no scalar number. The 3.122% is what we measured running their architecture on their data.*

WHAT WE CHANGED, AND WHY

The Architecture Upgrade

- **Change:** Replaced Bi-LSTM with an **MHA + Bi-Mamba-2*** hybrid.
- **Why:** Attention catches global symbol correlations instantly, while Mamba-2 fixes the Bi-LSTM's vanishing gradient memory bottleneck.

The Loss Function

- **Change:** Switched from MSE to **BCE Loss**.
- **Why:** MSE is for regression (leaving the model uncertain near 0.5).
  BCE is the standard for binary classification, forcing confident 0 or 1 bit decisions.

The Training Pipeline

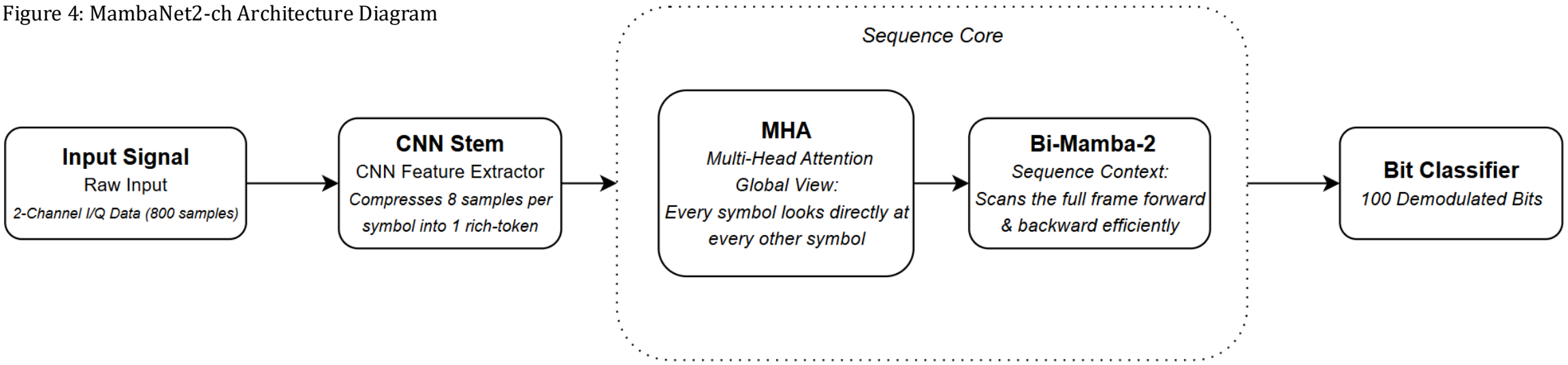
- **Change:** Added a **500K Synthetic Pretraining** stage.
- **Why:** Teaches the model core GMSK channel physics on simulated data *before* fine-tuning it on Zhu's 42K real, noisy samples.

We'll see what each change was actually worth in the ablation slide!

*MambAttention: Mamba with Multi-Head Attention for Generalizable Single-Channel Speech Enhancement. Kühne, N. L., et al. https://arxiv.org/abs/2507.00966

OUR MODEL: MambaNet-2ch DEMODULATOR

Figure 4: MambaNet2-ch Architecture Diagram



CNN compresses 8 samples/symbol into one token.
Attention then bidirectional Mamba refine each token using the whole frame. A linear head decides each bit.

MHA Block

- ❖ Reads all 100 tokens at once
- ❖ Captures global pairwise links
- ❖ "Which symbols depend on which?"

Bi-Mamba-2 Block

- ❖ Forward + Reverse selective scan
- ❖ Linear-time (Cheaper than MHA)
- ❖ "What does this bit's neighborhood tell me?"

~400K params · same budget as Zhu

TWO-STAGE TRAINING : PHYSICS FIRST, THEN REAL NOISE

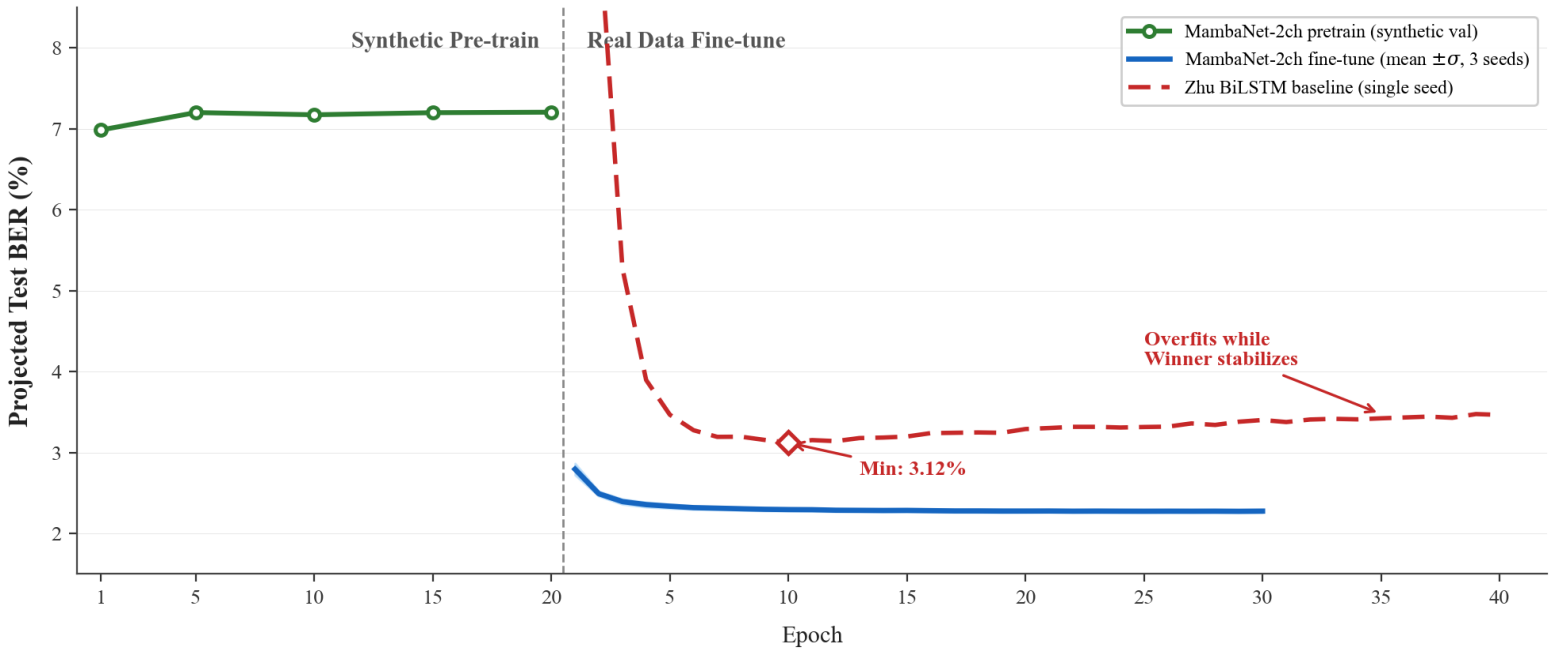


Figure 5: Pre-train and Fine-tune Transition

STAGE 1 : PRETRAIN

- 500K Synthetic GMSK Frames
- 20 Epochs
- Built our own GMSK + K channel Simulator

STAGE 2 : Fine-Tune

Zhu's 42K real samples

30 Epochs

Adapts to real channel statistics

❑ Why this works:

- ❖ Synthetic data is calibrated and unlimited
- ❖ Model learns clean GMSK Physics to Zhu's Real data
- ❖ Fine-Tune adapts that physics to Zhu's real data

"like a flight simulator before flying a real flight!"

OUR MODEL BEATS ZHU ON EVERY TEST CONDITION

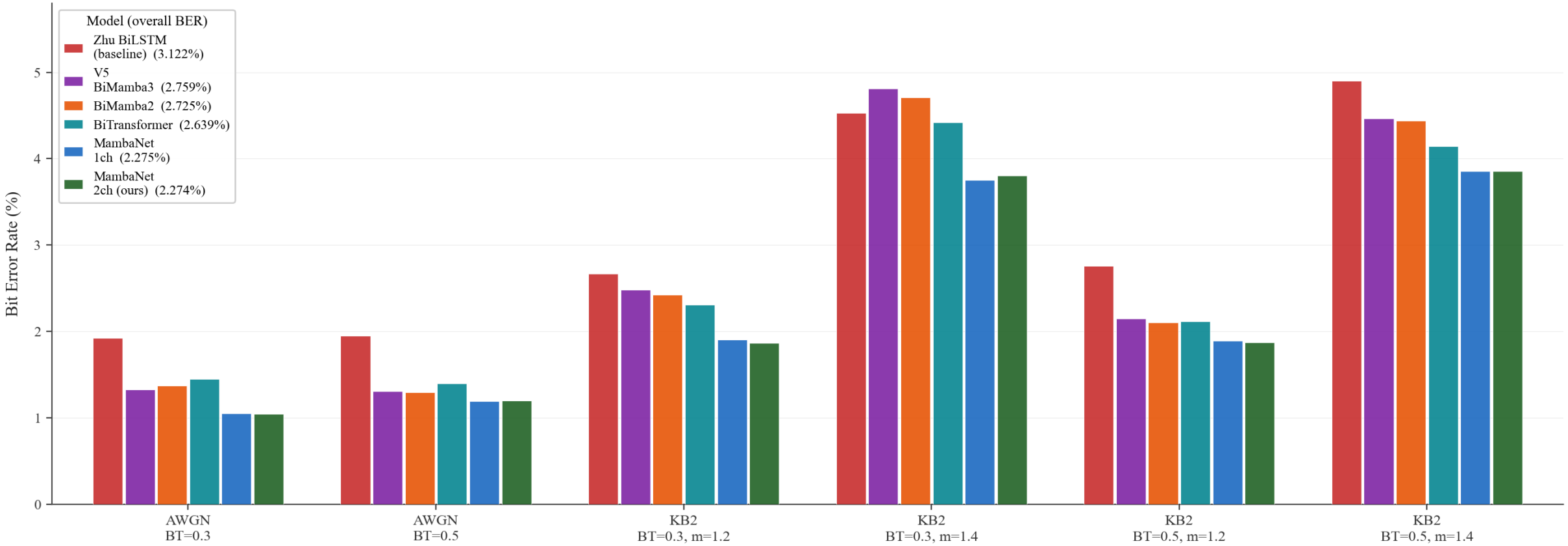


Figure 7: Model comparison across Different Model

2.275% vs 3.122% → *−27.2% relative*

We won on ***all 6 test conditions***, not cherry-picked
Paired t-test, n=6 conditions: **$p < 0.001$**

WHAT CONTRIBUTED: ABLATING EACH COMPONENT

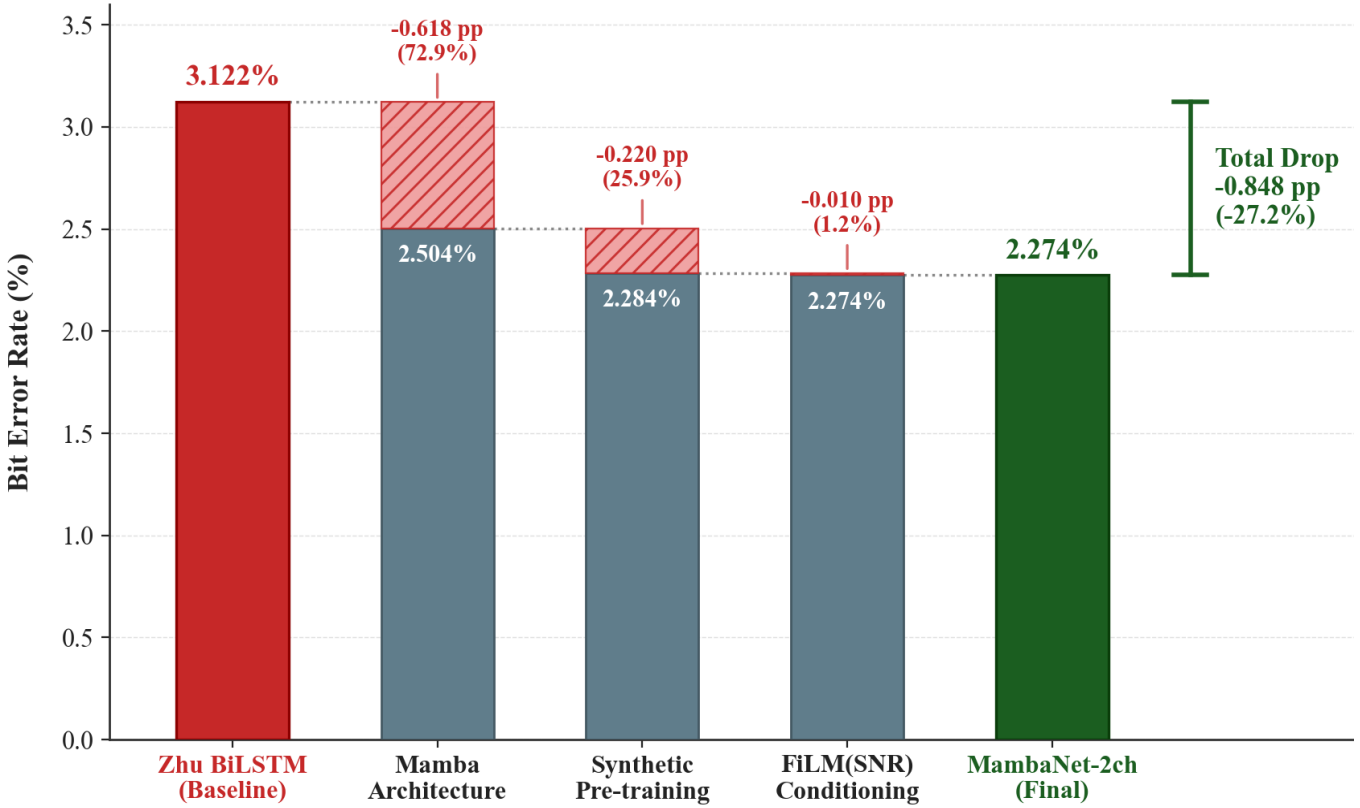


Figure 8: Ablation Study/ Architecture & Training Additions

- **BIGGEST DRIVER:**

Architecture Upgrade : MHA + Bi-Mamba2 *(-0.618pp, 73%)*

- **SECOND BIGGEST:**

Synthetic Pretrain *(-0.220pp, 26%)*

Not so surprising result:

FiLM (SNR) barely helps *(-0.010pp, 1%)*

We expected this to be a third pillar. Ablation said otherwise.

We tried a neural SNR estimator (V6) — also didn't move the needle.

Things we tried didn't work – and why

1. Test-Time Augmentation (Time-Reversing the Signal)

What we did: Fed the signal backwards during testing to average predictions.
Result: Catastrophic failure (+19.5% Error).
The Lesson: GMSK relies on differential phase encoding (bits depend on previous bits). Reversing the signal breaks the core physics of the encoding chain.

2. Viterbi Post-Processing

What we did: Added a classic algorithmic sequence decoder after our neural network.
Result: Literally zero effect (0.000% change).
The Lesson: This proves our Bi-Mamba-2 block already natively and perfectly learns the sequence constraints without needing algorithmic hand-holding.

4. Deep Neural SNR Estimator

- **What we did:** Built a complex neural network to estimate noise levels instead of a basic linear math function.
- **Result:** Statistically meaningless (+0.009% change).
- **The Lesson:** Adding deep learning complexity isn't always the answer. Our basic linear estimator was already extracting all available value.

5. The Wider Architecture Variant

- **What we did:** Scaled up the model width (d=192) to see if more capacity helped.
- **Result:** Numerically beat our final model
- (2.219% vs 2.275%).
- **The Lesson:** It failed the paired t-test (p=0.17). We rejected our best number to enforce strict statistical rigor. (It's not fair)

Conclusion : Receiver-side ML lowers the link budget bar

What We Achieved:

- **Baseline:** Reproduced Zhu 2023 1D-CNN + Bi-LSTM (3.122% BER)
- **Our Innovation:** Built MambaNet-2ch demodulator (2.275% BER)
- **The Result:** 27.1% relative reduction in bit errors ($p < 0.001$ across all 6 conditions)

Why It Matters (Electromagnetic Impact):

- **Effective Gain:** Provides an equivalent receiver gain of ~1 to 1.5 dB.
- **The Friis Tie-Back:** To hit the same BER target classically, you would need a larger G_t (bigger antennas) or a higher P_t (more transmit power).
- **Mass-Free Optimization:** Software-only improvement means zero added hardware mass or power draw on the deep-space probe.

The Takeaway: Receiver-side ML *complements* classical link engineering. It does not replace antenna design, FEC, or link margin- it simply lowers the baseline requirement.

Thank you. Questions?

REFERENCES

Machine Learning Architectures

- **Luan et al. (2026).** *MambaNet: Mamba-Assisted Channel Estimation Neural Network*. ICASSP 2026. *(Architectural blueprint for our MHA + BiMamba2 block)*
- **Dao & Gu (2024).** *Transformers are SSMs: Generalized Models and Efficient Algorithms*. arXiv:2405.21060. *(Core Mamba-2 SSM layer mechanics)*
- **Lahoti et al. (2026).** *Mamba-3: Improved Sequence Modeling using State Space Principles*. arXiv:2503.15569. *(Higher-order SSM variant built from source)*
- **Gu & Dao (2023).** *Mamba: Linear-Time Sequence Modeling*. arXiv:2312.00752.

Baseline & Signal Processing

- **Zhu et al. (2023).** *GMSK Demodulation Combining 1D-CNN and Bi-LSTM Network Over Strong Solar Wind Turbulence*. Radio Science. *(Primary baseline architecture and dataset)*
- **Murota & Hirade (1981).** *GMSK Modulation for Digital Mobile Radio Telephony*. IEEE Trans. Commun. *(GMSK BER theory and BT-product efficiency)*
- **K-Distribution Channel Modeling**
- **Jao (1984).** *Amplitude Distribution of Composite Terrain Scattered and Line-of-Sight Signal*. IEEE Trans. Antennas Propag. *(K-distribution envelope PDF formulations)*
- **Ward, Tough, & Watts (2006).** *Sea Clutter: Scattering, the K Distribution and Radar Performance*. IET. *(Complex K-distribution adapted for ionospheric simulation)*

REFERENCES

- **Tooling, Hardware & Datasets**
- **Hardware:** NVIDIA RTX 5070 (Blackwell architecture, sm_120 compute capability)
- **Environment:** Python 3.12.3 on Linux (WSL2)
- **Core Software:** PyTorch 2.11.0+cu130, CUDA 13.0, mamba-ssm 2.3.1 (built from source)
- **Key Dependencies:** causal-conv1d 1.6.1 (built from source), utilizing bf16 mixed precision
- **Test Set:** Zhu et al. Zenodo GMSK release (zenodo.org/record/5781913)

Reproducibility: Full PyTorch implementation, synthetic generators, and training logs available at: https://github.com/VaradhaCodes/Diffusion-Powered-Deep-Space-Signal-Recovery/

Thank you so much :P !